SUSHIL KUMAR YADAV

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PROFILE SUMMARY

Final-year Computer Science student specializing in AI/ML and IoT at GLA University, skilled in Python, Java, Machine Learning, and Generative AI, with strong foundations in the four core CS subjects: DSA, DBMS, OS, and CN.

EDUCATION

2023 – 2027	**B.Tech (CSE – AIML & IIOT),** GLA University, Mathura	CGPA: 6.8
2022 – 2023	**Intermediate (Class 12th),** Aditya Birla Intermediate College, Renukoot	75%
2020 – 2021	**High School (Class 10th),** Aditya Birla Intermediate College, Renukoot	79.3%

INTERNSHIP EXPERIENCE

Machine Learning Intern | Hindalco Industries Limited (Birla Soft)
May 2025 – Jun 2025

House Price Prediction

- Developed a house price prediction model using regression techniques**.**
- Performed data preprocessing, feature engineering, and model evaluation**.**
- Improved prediction accuracy through feature selection and hyperparameter tuning**.**

PROJECTS

Azithromycin Treatment Decision Support System
Python, Scikit-learn, Logistic Regression, Pandas, NumPy, Matplotlib

- Developed an ML-based clinical decision support system for pediatric watery diarrhea using 5,000 patient records and 11 input features.
- Built an end-to-end machine learning pipeline using Python and Scikit-learn, incorporating data preprocessing, feature scaling, one-hot encoding, stratified 80/20 train-test splitting, and Logistic Regression.
- Achieved **81.5% test accuracy and 0.74 ROC-AUC**, evaluating model performance using classification metrics, a confusion matrix, and ROC curve analysis.

AI-Powered Product Condition Verification for Safer E-Commerce
React, Fast API, YOLOv8, OpenCV, Tailwind CSS, SQL Alchemy

- Built an **AI-powered product verification system** using **YOLOv8** and computer vision to detect visible defects such as scratches cracks ,and dents**.**
- Developed an **image preprocessing and condition classification pipeline** to analyze product images and generate Good, Fair or Poor condition results**.**
- Integrated **Fast API APIs** with a React-based frontend to deliver automated verification results and improve **buyer trust and safer purchasing decisions.**

Touch-Free Gesture Control Application Using AI Hand Tracking
Python, OpenCV, MediaPipe, Computer Vision, AI

- Built an **AI-powered touch-free control system** using **MediaPipe and computer vision** to track hand movements and recognize predefined gestures in real time.
- Developed **a gesture recognition and mapping pipeline** to interpret gestures such as swipe, pinch, and open palm for controlling applications, presentations, and media.
- Enabled **low-cost, contactless interaction using a standard webcam**, improving accessibility, hygiene, and user experience without requiring specialized hardware.

SKILLS

Languages:	Java, Python, HTML, CSS, JavaScript, Node.js, React.js, Express.js
AI / ML:	PyTorch, LangChain, LangGraph, RAG Pipelines, Transfer Learning
Libraries / Frameworks:	TensorFlow, Scikit-learn, Pandas, NumPy, Seaborn, Matplotlib
Tools / Platforms:	Git, GitHub, Vercel, VS Code, Postman, Agile/Scrum, JIRA(Bug and Script Tracking)
Databases:	SQL, MongoDB